

Polarisation insensitive wavelength conversion of 10Gbit/s signals with SOAs in a Michelson interferometer

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Indexing terms: Semiconductor amplifiers, Optical frequency conversion

Polarisation insensitive wavelength conversion is demonstrated with a new configuration in which SOAs are placed in a Michelson interferometer. Wavelength conversion of 10Gbit/s signals is achieved with an improvement of the on/off ratio from 10 to 13dB, although the conversion is from shorter to longer wavelengths.

Introduction: Techniques to convert information from one wavelength to another will be needed in future WDM networks [1, 2]. The preferred features of such wavelength converters are transparency to bit rate as well as ability to convert high bit rate signals. Therefore, conversion techniques that preserve the signal in the optical domain have attracted much attention. To be practical, all-optical wavelength conversion needs to be polarisation insensitive and simple. Furthermore, high conversion efficiency is also necessary to allow converters to be cascaded. Considered techniques for all-optical wavelength conversion are optical modulation of lasers [3–5], four wave mixing in fibres and semiconductor optical amplifiers (SOAs) [6], as well as cross-phase and cross-gain modulation in SOAs [7, 8].

In this Letter we present a new and very efficient conversion technique, where SOAs are placed in a Michelson interferometer (MI) configuration. Penalty free wavelength conversion of 10Gbit/s data is demonstrated without the use of polarisation controllers.

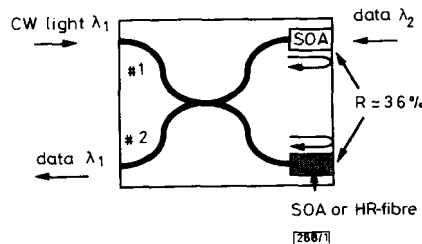


Fig. 1 Schematic diagram of wavelength converter: SOAs in Michelson interferometer

Principle of operation: The conversion principle reported here is based on the refractive index changes that occur in an SOA when an input signal depletes carriers due to stimulated recombinations. This refractive index change is used to modulate the phase for a second optical signal that is coupled through the SOA at the same time. To use this cross-phase modulation for wavelength conversion the SOA must be placed in an interferometric configuration. An attractive scheme is to place one SOA in each arm of a Michelson interferometer, because this configuration allows us in a simple way to change the phase in one arm relative to the other, as will be explained below. A schematic diagram of the converter is shown in Fig. 1 and the principle of operation is as follows: an SOA with one cleaved and one AR-coated facet is placed in each arm of a 3dB coupler. For a gain difference of 0 dB and a phase difference of π between the two arms, the CW signal at λ_1 coupled into arm 1 (see Fig. 1) will be reflected back into this arm. If, however, a second data signal at λ_2 is coupled into one of the SOAs as shown, an additional phase difference of π between the two interferometer arms results due to the refractive index change caused by the carrier depletion. As a consequence the CW signal will be reflected back into arm 2. Therefore, at the output of arm 2 the CW signal at λ_1 will be modulated according to the data input signal at λ_2 . It should be noted that an SOA in only one of the interferometer arms is sufficient if the other arm is terminated with an HR coating (see Fig. 1). Furthermore, the proposed SOA-MI configuration is insensitive to the polarisation of the incoming

data signal if the gain of the SOA is polarisation insensitive. In addition it can be fully integrated.

Experimental results: Wavelength conversion based on cross-phase modulation in an interferometric configuration has the advantage that the on/off ratio for the converted signal can be higher than that of the incoming signal, even when the conversion is performed from shorter to longer wavelengths. This is not possible for wavelength conversion based on cross-gain modulation [7]. To demonstrate the capability of the SOA-MI configuration, an SOA is placed in one arm of a 3dB fibre coupler whereas the fibre end in the other arm is supplied with an HR-coating as shown in Fig. 1. The SOA is based on the DCPBH structure [9] and is supplied with a double layer AR coating on one facet, while the other is uncoated giving facet reflectivities of 5×10^{-4} and 3.6×10^{-1} , respectively. The almost square active layer gives for a bias current of 35mA a polarisation insensitive double-pass gain of 16dB, which is sufficient to compensate for the round-trip loss in the amplifier arm (coupling loss between amplifier and fibre is 2×4 dB and facet loss is 5dB). The amplifier length is 600 μ m and the optical path difference between the two interferometer arms is less than 0.5mm. The phase difference between the two arms is adjusted by the fibre-amplifier coupling (the distance between the fibre and amplifier is varied) whereas the gain difference is controlled by the bias current to the amplifier. In a fully integrated version a passive phase section may be needed to adjust the phase difference.

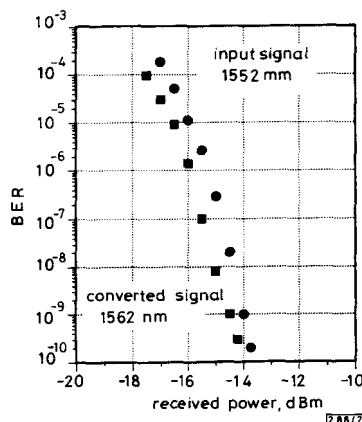


Fig. 2 Measured BER for wavelength conversion of 10Gbit/s signals (PRBS = 2^7-1)

With the SOA-MI configuration, a 10Gbit/s data signal at a wavelength of 1552nm (λ_2 in Fig. 1) is converted to a new wavelength of 1560nm (λ_1 in Fig. 1). The coupled power at the two wavelengths is 1 and 0.5mW, respectively. The measured bit-error-rate for the converted signal is shown in Fig. 2 together with that for the input signal detected directly without wavelength conversion. Although the conversion is from shorter to longer wavelengths, an improvement in the sensitivity for the converted signal is observed. The improvement is due to an increase in the on/off ratio from 10dB for the input signal to 13dB for the converted signal. It should be emphasised that no polarisation controllers are used for the incoming data signal.

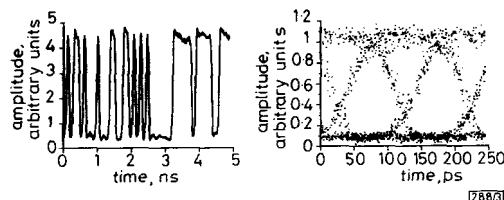


Fig. 3 Pulse pattern and eye diagram for converted signal at 10Gbit/s

A measured pulse sequence and an eye diagram for the converted signal are shown in Fig. 3. It should be noted that with the SOA-MI configuration the converted signal is not inverted compared to the input signal as is the case for conversion with cross-gain modulation. Moreover, because the converted signal passes the same 3dB coupler twice, the attainable on/off ratio is in principle insensitive to any wavelength dependency of the splitting ratio of the 3dB coupler.

Conclusion: High performance wavelength conversion of 10Gbit/s data is demonstrated with a new SOA-MI configuration. Because the investigated configuration is polarisation insensitive and can be fully integrated, it is considered very attractive for the implementation of a practical wavelength converter.

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Stabilisation of optical solitons by an acousto-optic modulator and filter

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Indexing terms: Soliton transmission, Acousto-optical devices, Optical communication

Continuously shifting the optical frequency by acousto-optic modulators may substantially improve the stability of a periodically amplified soliton transmission system with filtering. This method, which is analogous to the sliding filter technique, may also be applied to fibre loop soliton generation and storage.

Introduction: Recent experiments of long distance optical soliton propagation in a periodically amplified fibre system [1] have demonstrated that the soliton stability may be controlled by means of bandpass filters [2] and synchronous amplitude modulation. Another method to improve the soliton stability simply involves the continuous sliding of the filter bandpass along the transmission line [3,4]. However, this method cannot be applied to recirculating soliton fibre loop memories and lasers.

The present analysis shows that a substantial improvement in the soliton stability in a periodically amplified system may be achieved by inserting a travelling-wave acousto-optic modulator and a bandpass filter after each amplifier. This leads to a continuous wavelength shift along the link. Note that frequency shifting and filtering could be obtained with a single integrated optics device (the acousto-optic modulator and filter, AOMF) that exploits the acoustically phasematched TE-TM conversion in a waveguide [5].

Theory: In dimensionless soliton units [3,6], the action of a bandpass filter may be described by the transfer function in the frequency domain $H(\omega - \omega_0) = (1 + 2i(\omega - \omega_0)/B)^{-1}$, where ω_0 is the centre frequency of the filter bandpass and B is its bandwidth. The action of the AOM may be represented by the transfer function in the time domain $S(T) = \exp[i\alpha T + i\theta - \gamma]$, where α is the acoustic drive frequency, θ is a constant phase and γ is a loss coefficient

owing to the limited AOM diffraction efficiency.

Whenever the length of the loop, say, z_0 is much shorter than the dispersion length $z_0 = t_0^2/|\alpha|$ (α is the group velocity dispersion and t_0 is the pulse time width), we may average along the propagation co-ordinate the lumped amplifier gain, filtering [6], frequency shifting, and obtain

$$\partial_Z V - \frac{i}{2} \partial_{TT} V - i|V|^2 V - i\alpha_0 TV = \delta V + V_f \partial_T V + \beta \partial_{TT} V \quad (1)$$

Here $\delta = Gz_0/z_a - \gamma z_0/z_a - \Gamma > 0$ is the excess gain, Γ is the distributed fibre loss coefficient, $\exp\{2G\}$ is the power gain factor at each amplifier, $\beta = 2z_0/(B^2 z_a)$, $V_f = 2z_0/(z_a B)$, and $\alpha_0 = \alpha z_0/z_a$. The formal equivalence of the present scheme with the sliding filter method is proved by the change of variables

$$Q(T - \alpha_0 Z^2/2, Z) = V(T, Z) \exp[-i\alpha_0 ZT - i\alpha_0 Z^2/B - i\alpha_0^2 Z^3/6] \quad (2)$$

We obtain from eqns. 1 and 2 the new equation for Q :

$$\partial_Z Q - \frac{i}{2} \partial_{\tau\tau} Q - i|Q|^2 Q = \delta Q + \beta(\partial_\tau + i\omega_f(Z))^2 Q \quad (3)$$

where $\tau = T - \alpha_0 Z^2/2 + V_f Z$, and $\omega_f(Z) = \alpha_0 Z$. Eqn. 3 describes a transmission system where the filters linearly slide with Z at the rate α_0 [3].

Without the filter (i.e. with $\delta = \beta = V_f = 0$), the AOM leads to the continuous shift with Z of the centre frequency of the solitons. In fact, the one soliton solution of eqn. 1 reads

$$V(\tau, Z) = \eta \operatorname{sech}[\eta(\tau - \xi(Z))] \exp[i\kappa(Z)\tau + i\psi(Z)] \quad (4)$$

Here the soliton position $\xi = \alpha_0 Z^2/2 + \kappa_0 Z$, frequency $\kappa = d\xi/dZ$, and phase $d\psi/dZ = (\eta^2 - \kappa^2)/2$. With the filter, we obtain from first order soliton perturbation theory that the soliton frequency remains trapped if $|\alpha_0| < \alpha_c = 8\delta\sqrt{(\delta)/(3\sqrt{3}\beta)}$. In this case, the soliton pulse represents a stable balance between the filter loss and the excess gain δ . In fact, for large Z the soliton frequency approaches $\kappa = 3\alpha_0/(4\beta\eta^2)$, whereas $\eta = 1$ for $\delta = \beta/(1/3 + \kappa^2)$ [7]. On the other hand, low intensity radiation and noise do not remain trapped by the filter, but are continuously shifted by the AOM and are rapidly absorbed. The suppression of the interac-